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Course:

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ASSIGNMENT NO # 01

Code:

```
ceaser_cipher.py 1 431567.py 1 Untitled-1.py 5
C: > Users > Laptop > Downloads > Untitled-1.py > ...
1  # Caesar Cipher
2  def caesar_encrypt(text, shift):
3      encrypted_text = ""
4      for char in text:
5          # Check character cases
6          if char.isupper():
7              # range of uppercase letters (A-Z)
8              encrypted_char = chr((ord(char) - ord('A') + shift) % 26 + ord('A'))
9              encrypted_text += encrypted_char #uppercase
10         elif char.islower():
11             # range of lowercase letters (a-z)
12             encrypted_char = chr((ord(char) - ord('a') + shift) % 26 + ord('a'))
13             encrypted_text += encrypted_char #lowercase
14         # If not a letter, keep it unchanged
15         else:
16             encrypted_text += char
17     return encrypted_text
18 # decrypt the ciphertext
19 def caesar_decrypt(ciphertext, shift):
20     decrypted_text = ""
21     for char in ciphertext:
22         # uppercase letter
23         if char.isupper():
24             decrypted_char = chr((ord(char) - ord('A') - shift) % 26 + ord('A'))
25             decrypted_text += decrypted_char
26         # lowercase letter
27         elif char.islower():
28             decrypted_char = chr((ord(char) - ord('a') - shift) % 26 + ord('a'))
29             decrypted_text += decrypted_char
30         # If not a letter, keep it unchanged
```

```
ceaser_cipher.py 1 431567.py 1 Untitled-1.py 5
C: > Users > Laptop > Downloads > Untitled-1.py > ...
19 def caesar_decrypt(ciphertext, shift):
20     # If not a letter, keep it unchanged
21     else:
22         decrypted_text += char
23     return decrypted_text
24 #Main Program
25 # Taking input from user
26 message = input("Enter your message: ")
27 shift_value = int(input("Enter shift value: "))
28 # Encrypting the message
29 encrypted = caesar_encrypt(message, shift_value)
30 print("\nEncrypted Message:", encrypted)
31 # Decrypting the message
32 decrypted = caesar_decrypt(encrypted, shift_value)
33 print("Decrypted Message:", decrypted)
```

Output:

```
61 shift_value = int(input("Enter shift value: "))
62
PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 7 Python + - [ ] [ ] [ ]
PS C:\Users\Laptop> & C:/Users/Laptop/.local/bin/python3.14.exe c:/Users/Laptop/Downloads/Untitled-1.py
Enter your message: all services are okay?
Enter shift value: 131

Encrypted Message: bmm tfswjdft bsf plbz?
Decrypted Message: all services are okay?
PS C:\Users\Laptop>
PS C:\Users\Laptop> 14
14
PS C:\Users\Laptop> |
```

Spaces: 4 UTF-8 { Python Signed out 3.14.3 Python 3.14

Record high 17:04 28-Feb-26

EXPLANATION:

Encryption Function

```
def caesar_encrypt(text, shift):
```

This function takes original message (text) and shift value and returns encrypted text.

```
    encrypted_text = ""
```

An empty string is created to store encrypted result.

```
    for char in text:
```

Loop is used to check each character one by one.

```
        if char.isupper():
```

If character is uppercase letter (A–Z), it is shifted.

```
            chr((ord(char) - ord('A') + shift) % 26 + ord('A'))
```

- `ord()` converts letter into ASCII number.
- Shift value is added.
- `% 26` is used for wrap-around (Z → A).
- `chr()` converts number back to character.

```
        elif char.islower():
```

Same process is applied for lowercase letters (a–z).

```
else:  
    encrypted_text += char
```

Spaces and special characters are not changed.

```
return encrypted_text
```

Returns encrypted message.

Decryption Function

```
def caesar_decrypt(ciphertext, shift):
```

This function converts encrypted text back to original message.

The only difference is that shift value is subtracted:

```
(ord(char) - ord('A') - shift) % 26
```

This moves letters back to their original position.

Main Program

```
message = input("Enter your message: ")  
shift_value = int(input("Enter shift value: "))
```

User enters message and shift value.

```
encrypted = caesar_encrypt(message, shift_value)
```

Message is encrypted.

```
decrypted = caesar_decrypt(encrypted, shift_value)
```

Encrypted message is decrypted again.

Security Analysis

Strengths

- Easy to implement.
- Good for learning basic cryptography.
- Fast execution.

Weaknesses

- Only 25 possible shift keys.
- Easy to break using brute force attack.

- Vulnerable to frequency analysis.
- Not secure for modern security systems.

Conclusion

In this program, Caesar Cipher is successfully implemented.
The program preserves uppercase, lowercase, and special characters.

However, Caesar Cipher is a weak encryption method and not suitable for modern secure communication.